FIKR Model — Lesson Plan

Learning Leading to Production

Title	Exponential and Logarithmic Equations and Inequalities	**Subject / Unit**	Topic 6 — Exponential and Logarithmic Functions · Lesson 6-6
Grade	Grade 11 — Algebra II	**Date**	Week 7 · Semester 1, 2026–27
Format	[ ] 40-minute [X] 60-minute	**Teacher**	Mr Malek Thiab

Essential Question (curriculum map)

How can we solve exponential and logarithmic equations and inequalities?

Lesson Objectives (curriculum map — verbatim)

1. Solve exponential equations and inequalities by applying the concept of a common base.
2. Solve logarithmic equations and inequalities by using the rules of logarithms.
3. Use logarithms to solve exponential equations and models.

Vocabulary (curriculum map) · Assessments (curriculum map)

Vocabulary: exponential equation ; exponential inequality ; logarithmic equation ; logarithmic inequality.

Assessments: “Give it a go” questions during practice ; “Time to Check” at the Mastery Gate.

0 Preparation Before Class

Learning Outcome: What will the student be able to do?

Students will solve exponential equations and inequalities by writing both sides with a common base, and explain why the direction of an inequality flips when the base is between 0 and 1 (objective 1; HSA.SSE.A.2).

Students will solve logarithmic equations and inequalities using the rules of logarithms, and reject extraneous solutions by checking every argument against the domain (objective 2; HSA.CED.A.1).

Students will use logarithms to solve exponential equations and models when the bases will not match (objective 3; HSF.LE.A.4).

Students will explain each step as following from the line before it, and justify the solution method (HSA.REI.A.1; MP.2 and MP.3).

If time runs short, cut Smart Production and carry it into the next session — never the Mastery Gate, which is the evidence.

Pre-Class Preparation — Topic/Link and Evidence of Readiness: What will the student complete before class?

Flipped video (4 min): “Same base, or a logarithm?” — rewriting a number as a power, and deciding when a logarithm is needed. Posted on the LMS 48 hours ahead.

Two embedded EdPuzzle questions: rewrite 27 as a power of 3, and evaluate $\log_2 32$.
Evidence of readiness: EdPuzzle completion report plus both embedded answers submitted before the bell.

1 Diagnostic 6 min

Tool Used: What evidence indicates differences in students' current levels?

3-question Quizizz (≤ 4 minutes) — slide 4: solve $5^x = 125$; write 16^x as a power of 2; evaluate $\log_3 81$ and $\log_2 \frac{1}{8}$. No calculator.

Cross-referenced against the overnight EdPuzzle watch analytics.

Answers: $x = 3$; 2^{4x} ; 4 and −3 — the first two test common bases and power of a power, the third tests the meaning of a logarithm.

Where is the learning gap? What will you adjust in your instruction accordingly?

Expected gap 1 — writing both sides with one base (power of a power, negative exponents); if Q1 or Q2 fail, the lesson stalls at block 1.

Expected gap 2 — evaluating a logarithm with a negative result; this is the entry to logarithmic equations.

Routing: 0–1 correct → re-teach block 1 with the teacher. 2 → straight to the examples. 3 → begin the Investigate prompt immediately.

2 Targeted Instruction 18 min

Required Instruction and Modeling: What needs direct explanation? Which example will demonstrate the thinking process and the expected quality standard?

Slide 5 — Exponential equations (objective 1). Match the bases, then equate the exponents: $2^{x+1} = 32$ becomes $x + 1 = 5$; $9^x = 27^{x-1}$ becomes $2x = 3x - 3$, so $x = 3$. The step is justified because an exponential function is one-to-one (HSA.REI.A.1).

Slide 6 — Exponential inequalities (objective 1). With a base above 1 the direction stays; with a base between 0 and 1 it flips: $\left(\frac{1}{2}\right)^x < \frac{1}{16}$ gives $x > 4$. Test one value inside and one outside every answer.

Slide 7 — Logarithmic equations and inequalities (objective 2). Use the rules, convert to exponential form, then check the domain: $\log_2(x+3) + \log_2(x-3) = 4$ gives $x = 5$ and the extraneous $x = -5$. $\log_2(x-1) < 3$ needs two conditions and gives $1 < x < 9$.

Slide 8 — Logarithms for exponential equations and models (objective 3; HSF.LE.A.4). When the bases will not match, take a logarithm: $1000e^{0.05t} = 3000$ gives $t = \dfrac{\ln 3}{0.05} \approx 22.0$ years.

A NEW example set — it does not repeat the flipped video.

Quick Check for Understanding: What question or evidence confirms that students are ready to move on to practice?

On whiteboards, 60 seconds: solve $4^{x-3} = 16$.

Expected: $x = 5$ ($16 = 4^2$, so $x - 3 = 2$). Watch for $x = 4$ (dividing 16 by 4) and $x = 1$.

80% correct → release guided practice. Below 80% → one more modelled example with a different base.

3 Practice 12 min

Guided Task: How will the student progress from guided to independent practice? What feedback will the student use?

Two "Give it a go" problems, identical for every student, worked together with the teacher:

(1) Solve $3^{x-2} = 81$. (2) Solve $\log_4(x + 2) = 3$.

Answers: (1) $81 = 3^4$, so $x - 2 = 4$ and $x = 6$. (2) $x + 2 = 4^3 = 64$, so $x = 62$; check that $x + 2$ is positive.

Feedback: worked live on the board step by step; students self-mark and circle their OWN first error.

Independent Task and Feedback Tool: students choose their route and may switch at any time

Delivered through the "Choose Your Route" sheet with live feedback in Pear Deck. Routes are named for the task, not for the student.

PRACTICE — Secure the method. Common-base equations, one inequality, one logarithmic equation and one equation solved with a logarithm.

APPLY — Use it in context. An inequality with a base below 1, a logarithmic equation with an extraneous solution, and a growth model solved with a logarithm (a Jeddah internet-subscriber model, model values only).

INVESTIGATE — Find out why. An error analysis of a logarithmic equation, why the same inequality flips with base 0.5 and not with base 2, and an equation that cannot be solved by matching bases. The "Done when…" criteria for all three routes are on the sheet and in the Differentiation box below.

Every student sits the same Mastery Gate. The gate is not differentiated, because it is the evidence.

4 Production — Stage One 12 min

Production Task: What new task, situation, or product will allow the student to demonstrate learning?

Non-routine transfer task — this context has not appeared in practice (slide 10).

A model for two Red Sea resorts (model values, not forecasts): $A(t) = 40\,000\,(1.25)^t$ and $B(t) = 20\,000 \cdot 2^{t/3}$ visitors per year, t years after opening.

(1) Solve $A(t) = 125\,000$ with a logarithm. (2) Solve $A(t) > 200\,000$ and give whole years. (3) Solve $B(t) = 160\,000$ by matching bases, then $A(t) = 160\,000$, and say which resort gets there first.

Answers: (1) $(1.25)^t = 3.125$, $t = \dfrac{\ln 3.125}{\ln 1.25} \approx 5.1$ years. (2) $(1.25)^t > 5$, $t > 7.2$, so 8 whole years (the base is above 1, so the direction stays). (3) Resort B: $2^{t/3} = 8$ gives $t = 9$ years; Resort A: $(1.25)^t = 4$ gives $t \approx 6.2$ years, so Resort A is first.

Students then hand their reasoning to the AI critic and ask it to challenge any step they have not justified.

Product Quality Criteria and the Role of AI: What is the minimum acceptable standard? What must the student complete independently?

Minimum acceptable standard: the method named (match bases or take a logarithm) before the calculation, exact expressions before decimals, and units (years, visitors) in every answer.

Role of AI: critic only. Briefed prompt — "Challenge my reasoning and point out any step I have not justified." Never "solve this".

Completed independently by the student: the written justification of each step. AI may challenge it; it may not write it.

5 Presentation — Evidence of Learning 6 min

Evidence of Learning and Success Criterion: What will the student present or explain, and how? What must be clearly demonstrated for learning to be considered achieved?

"Time to Check" — one individual task. No notes, no partner, no AI.

1. Solve $4^{x+1} = 8^x$. 2. Solve $\log_3 x + \log_3(x+6) = 3$. 3. Solve $\left(\frac{1}{2}\right)^x > 8$.

Answers: 1. $x = 2$. 2. domain $x > 0$, candidates 3 and −9, so $x = 3$. 3. $x < -3$ (the base is below 1, so the direction flips).

Done when: answers are exact, the rejected value in question 2 is named, and the direction in question 3 is justified.

Scored live in Pear Deck as submissions arrive; result logged in the Impact Dashboard Mastery Rate.

6 Final Product — Refinement 6 min

Refinement and Final Product: What revision will the student make based on feedback? What will the final product look like?

Students revise using the AI critique and the teacher's micro-conference note. Final product, three equally acceptable formats: a Canva one-pager, a photographed hand-written solution, or a 45-second voice note.

Required content: the Red Sea resort solution with the method named at each step; ONE worked non-example — keeping an extraneous solution, or not flipping an inequality with a base below 1 — with a sentence on what goes wrong; and a caption on how a planner would use the answer.

Graded on originality and clarity of reasoning, not correctness alone.

Presentation/Publication and Reflection: Where will the product be presented or published? What reflection question will close the learning cycle?

Published to the class LMS portfolio; the strongest go on the Mathematics Department wall as the Topic 6 modelling set.

Reflection question: "Which decision is harder for you — matching the bases or choosing a logarithm, and how do you know which one a problem needs?"

Total minutes across the seven stages: 60 (6 + 18 + 12 + 12 + 6 + 6) — matches the 60-minute format, including transitions.

★ Required Pillars

Differentiation · Adaptive Learning · Saudi Connection · Exam Alignment

Differentiation — three routes named for the task; students choose, and may switch at any time

PRACTICE — Secure the method. Common-base equations, one inequality, two logarithm-based problems. Done when: both sides share one base before the exponents are equated, and each logarithm answer is checked.

APPLY — Use it in context. A base-below-1 inequality, an extraneous solution, and a growth model solved with a logarithm. Done when: the direction is justified by the base, and each solution is tested against the domain.

INVESTIGATE — Find out why. An error analysis, why the direction depends on the base, and an equation that cannot be matched. Done when: each explanation uses the domain or the base.

No student is labelled. The routes describe tasks; every student sits the same Mastery Gate.

Adaptive Learning — how the lesson branches on evidence

Entry branch — the Quizizz result routes each student to the re-teach table, straight to the new content, or the Investigate prompt.

Mid-lesson branch — Pear Deck flags error patterns during independent practice; flagged students get a 2-minute micro-conference while others continue.

Exit branch — the Mastery Gate routes each student: PASS → Enrichment & Challenge; NOT YET → Targeted Learning Clinic at the start of the next lesson.

Routing is communicated privately through the LMS, never read aloud.

Saudi Culture & Real-World Connection

Two Red Sea resorts (Vision 2030 tourism; model values only) with different growth patterns; the A3 project adds Phase 2 planning and a falling water-use model (Vision 2030 sustainability).

SAT / SAAT (Tahsili) / GAT (Qudurat) Connection

SAT — Advanced Math: $9^{x-1} = 27^{x/2}$, find x. Convert 9 and 27 to base 3, equate the exponents, $x = 4$. Name the trap: equating exponents before the bases match.

GAT (Qudurat) — quantitative comparison: A is the value of a with $2^a = 8$; B is the value of b with $4^b = 8$. $a = 3$ and $b = \frac{3}{2}$, so A is greater. Name the trap: assuming equal right-hand sides give equal exponents.

SAAT (Tahsili) — $\log_2 x + \log_2(x-1) = 1$: $x(x-1) = 2$ gives candidates 2 and −1; the domain needs $x > 1$, so $x = 2$. Name the trap: keeping the extraneous solution.

Suggestions for Improvement: